

Final Report on

**Video Streaming Web App with
ML-Powered Recommendation and AI
Integration**



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Abstract

Our project focuses on developing a web app video streaming application that utilizes machine learning to enhance user experience through personalized content recommendations. The system leverages both collaborative filtering and content-based filtering to analyze user preferences, watch history, and interaction patterns, enabling intelligent video recommendations tailored to individual viewing habits.

One of the major features of the application is its differentiated access model, where a user can both create and upload the video as well as view the videos uploaded by other users. This allows the user to be the content creator as well as a viewer.

With Artificial Intelligence integration for the creator, it helps to generate the thumbnails, video description as well as eye-catching titles.

Also, with the help of Machine Learning the recommendation system continuously adapts to user behavior, ensuring the viewer a tailored experience and adapts to the user liking and history which brings viewer friendliness.

Keywords: Web App, Video Platform, Machine Learning, Smart Recommendations, User Behavior, AI Features, Creator Tools, Viewer Experience.

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Yours sincerely,

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Abbreviations

ABR	Adaptive Bitrate Streaming
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
CDN	Content Delivery Network
CSS	Cascading Style Sheets
DB	Database
Drizzle ORM	Drizzle Object Relational Mapping
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
JS	JavaScript
JSON	JavaScript Object Notation
MERN	MongoDB, Express.js, React, Node.js
ML	Machine Learning
Next.js	Next JavaScript Framework
NLP	Natural Language Processing
ORM	Object-Relational Mapping
REST	Representational State Transfer
SQL	Structured Query Language
tRPC	TypeScript Remote Procedure Call
TS	TypeScript
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
VOD	Video on Demand

Chapter 1

Introduction

1.1 Overview

In today's content-rich digital world, discovering videos that match user interests can often feel overwhelming. To tackle this challenge, we introduce NepTube, a web-based video streaming platform designed to deliver a personalized and engaging viewing experience through the power of artificial intelligence and machine learning.

At the heart of NepTube lies an intelligent recommendation system powered by machine learning algorithms, including collaborative filtering [1] and content-based filtering. These systems analyze user preferences, watch history, and interaction patterns to generate highly relevant video suggestions that adapt over time [2].

Going beyond recommendations, NepTube also integrates AI-powered tools for content creators, including automatic thumbnail generation, video title suggestions, and description drafting [3], helping creators make their content more appealing and discoverable with minimal effort.

While the platform draws inspiration from larger services like YouTube [4], NepTube focuses on a simpler, more focused user experience. Core features include the ability to create playlists, like or dislike videos, comment, and subscribe to favorite creators—fostering community and interaction.

By combining adaptive machine learning with creator-focused AI tools and essential social features, NepTube aims to create a user-friendly streaming environment that supports both content discovery and creation.

1.2 Problem Statement

In Nepal, the growing demand for digital video content faces several challenges that limit the overall user experience and content creator opportunities on streaming platforms. Despite improvements in internet access and usage, the following issues remain prevalent:

- **Heavy Reliance on Foreign Platforms:** Most Nepali users depend on international streaming services, which often do not prioritize local languages, culture, or content preferences.
- **Lack of Localized Video Streaming Platforms:** There is a shortage of streaming services designed specifically for the Nepali market, which limits access to culturally relevant content and fails to address local user needs.
- **Content Discovery Overload:** Users in Nepal often feel overwhelmed by the vast number of available videos, making it difficult to find content that truly matches their interests.
- **Monetization Challenges:** Content creators face difficulties in monetizing their work effectively, impacting the sustainability of local content production.

This project aims to develop NepTube, a web-based video streaming platform designed specifically for the Nepali audience [5]. Leveraging machine learning, the platform will provide personalized video recommendations based on users' viewing habits and preferences. Alongside an intuitive user experience, NepTube will offer essential social features such as playlists, comments, likes/dislikes, and subscriptions to foster community engagement. Additionally, AI-powered tools [6] will assist content creators by automatically generating video thumbnails, titles, and descriptions to enhance content visibility. By focusing on localization, scalability, and creator support, NepTube seeks to improve content discovery, promote Nepali creators, and contribute to the growth of Nepal's digital entertainment landscape.

1.3 Objectives

Following are the main objectives of our project:

- Create a full-fledged online streaming platform where users will be able to smoothly watch videos, choose between different quality levels [7], and enjoy stable playback even on slower networks.
- Build a system that lets creators upload their content easily, manage their profiles, and customize things like thumbnails, titles, and video details with AI helpers [8].
- Add features that bring life to the platform, like playlists, comments, likes, and channel subscriptions.

- Research popular recommendation techniques [9] including collaborative filtering and content-based filtering, drawing comparisons between how the two work before implementing which fits best with our platform.
- Integrate ad content naturally inside the video player.

1.4 Aim

The aim of this project is to develop NepTube, a scalable and user-friendly web-based video streaming platform specifically designed for the Nepali audience. The platform will leverage advanced machine learning [10, 11] techniques to provide personalized video recommendations that reflect individual user preferences and local cultural relevance.

Additionally, NepTube will integrate AI-powered tools to assist Nepali content creators in enhancing their videos through automatic thumbnail generation, title suggestions, and description drafting. By incorporating interactive features such as playlists, comments, likes/dislikes, and subscriptions, the platform aims to foster a vibrant online community that encourages user engagement and supports local creators. Overall, this project seeks to bridge the gap in the digital entertainment ecosystem in Nepal by offering a localized, intelligent, and engaging video streaming experience tailored to the unique needs of Nepali viewers and creators contributing in the entertainment sector ecosystem.

1.5 Motivation

Nepal's digital landscape is evolving rapidly, with increasing internet accessibility playing a pivotal role in transforming content consumption habits. However, most Nepali users currently rely on international video streaming platforms that often overlook local cultural contexts, linguistic diversity, and the unique preferences of Nepal's audience. This gap limits the availability of relevant, engaging, and culturally resonant content tailored specifically for Nepali viewers.

Simultaneously, local content creators face significant challenges in expanding their reach and enhancing their content's appeal due to the lack of advanced, user-friendly tools. The absence of AI-driven features for content optimization and personalized discovery constrains their growth and limits user engagement.

Driven by these challenges, NepTube aims to develop a web-based, localized video streaming platform that leverages cutting-edge machine learning techniques [12] to deliver per-

sonalized recommendations and AI-powered creator tools. This project is motivated by the vision to empower Nepali creators, enhance content discovery, and foster a vibrant, interactive community. NepTube aspires to bridge the gap between advanced technology and local cultural relevance, thereby enriching Nepal's digital entertainment ecosystem and offering a truly tailored viewing experience.

1.6 Expected Outcomes

The expected outcomes of this project are as follows:

- A fully functional web-based video streaming platform accessible from any modern browser.
- An intelligent recommendation system that delivers personalized video suggestions based on user behavior and preferences.
- AI-powered creator tools for automatic thumbnail generation, title suggestions, and video description drafting.
- A robust content moderation pipeline integrating AI-based toxicity detection and sentiment analysis.
- Secure user authentication with role-based access control (viewer, creator, admin).
- A subscription management system with tiered premium features and local payment gateway integration.
- Comprehensive admin and creator dashboards for analytics, content management, and platform monitoring.
- A scalable and deployable platform architecture using modern serverless technologies.

1.7 Scope and Applications

1.7.1 Scope

- **Development of a Web-Based Platform:** Build a responsive and user-friendly video streaming web application accessible via standard web browsers.

- **User Interaction Features:** Implement core functionalities such as video upload, viewing, commenting, liking/disliking, playlist creation, and channel subscription.
- **Personalized Recommendation System:** Integrate machine learning algorithms (collaborative and content-based filtering) to provide customized video suggestions based on user activity and preferences.
- **AI-Powered Creator Tools:** Provide content creators with artificial intelligence tools for automatic thumbnail generation, title suggestions, and video descriptions.
- **Support for Local Content:** Focus on promoting Nepali videos and regional content to support local creators and cater to the cultural preferences of Nepali users.
- **User Authentication and Access Control:** Enable secure sign-up, login, and role-based access (viewer/creator) to manage content uploads and interactions.
- **Scalable Architecture:** Design the system to handle increasing numbers of users and growing video content efficiently over time.
- **Community Engagement:** Encourage active participation through comment sections, feedback mechanisms, and content subscriptions.
- **Cross-Platform Accessibility:** Ensure the platform is optimized for use across desktops, laptops, and other devices with internet browsers, without reliance on mobile-specific apps.

1.7.2 Applications

- **Video Streaming Platform:** Allows users to watch, upload, and share video content through a centralized web application.
- **Content Recommendation:** Provides personalized video suggestions using machine learning to enhance user engagement and retention.
- **Digital Content Creation Support:** Offers AI-based tools to assist creators in generating thumbnails, titles, and descriptions for their videos.
- **Educational Use:** Can be adapted for sharing educational videos, tutorials, and online learning content for students and teachers.
- **Community Engagement:** Enables user interaction through comments, likes, and subscriptions, fostering a sense of online community.

- **Marketing and Branding:** Businesses and creators can use the platform to reach target audiences by uploading promotional and informational videos.
- **Entertainment Hub:** Serves as a go-to destination for users seeking diverse video content, from music and vlogs to documentaries and entertainment clips.

1.8 Feasibility Study

1.8.1 Technical Feasibility

The development of NepTube is technically feasible using currently available and accessible web technologies. The platform will be built using modern web development tools such as Next.js [13], TypeScript [14], and React [15]. Machine learning models for video recommendation can be implemented using popular open-source libraries and AI APIs [16]. AI features for generating thumbnails, titles, and descriptions can be added using pre-trained models such as Whisper v3 for transcription [17]. Since the project is web-based, it can be accessed from any device with a browser, eliminating the need for mobile app development. The technical requirements are within the skill set of the development team, making implementation realistic and manageable.

1.8.2 Economic Feasibility

The development of NepTube is economically practical, especially within the scope of a student or academic project. Most of the tools and technologies required—such as web development frameworks, machine learning libraries, and database systems—are freely available as open-source resources. Initial costs such as domain registration and basic cloud hosting can be managed using free or low-cost service tiers. Since the project does not require expensive hardware, paid APIs, or large-scale infrastructure during its initial phase, the overall financial requirements are minimal. This makes the project viable within limited budgets, while still allowing for future expansion if the platform gains traction.

1.8.3 Operational Feasibility

The NepTube platform is operationally feasible as it targets users already familiar with video streaming interfaces. The application follows established UX patterns from popular platforms, minimizing the learning curve. Content creators can easily upload and manage their videos through an intuitive dashboard, while viewers navigate the platform using

familiar browsing, search, and recommendation features. System administration is streamlined through the admin dashboard with AI-assisted moderation tools. The platform requires minimal ongoing maintenance due to its serverless architecture deployed on Vercel's hosting infrastructure.

1.9 System Requirements

1.9.1 Software Requirements

- **Operating System:** Windows 10/11, macOS 12+, or Linux (Ubuntu 22.04+)
- **Runtime:** Bun 1.x or Node.js 20+
- **Package Manager:** Bun (bundled with runtime)
- **Database:** PostgreSQL 16.x (hosted on Neon serverless)
- **Version Control:** Git 2.x+
- **IDE:** Visual Studio Code (recommended)
- **Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari (latest versions)

1.9.2 Hardware Requirements

- **Processor:** Intel Core i5 or equivalent (minimum), Intel Core i7/AMD Ryzen 7 (recommended)
- **RAM:** 8 GB (minimum), 16 GB (recommended)
- **Storage:** 500 MB for application code, 10 GB+ for video storage
- **Network:** Broadband internet connection (minimum 10 Mbps)
- **Display:** 1920×1080 resolution (recommended)

1.10 Project Schedule

The development of NepTube followed a structured schedule divided into major phases. The project was completed over a period of approximately 8 months.

Table 1.1: *Project Schedule*

Phase	Activity	Duration	Deliverable
1	Requirement Analysis & Planning	Month 1–2	SRS Document, Project Plan
2	System Design	Month 2–3	Flowcharts, Use Cases, Sequence Diagrams
3	Core Development	Month 3–5	Authentication, Video Upload, Streaming
4	AI Integration	Month 5–6	Recommendations, Moderation, Content Generation
5	Premium & Community Features	Month 6–7	Payments, Subscriptions, Community Posts
6	Testing & Documentation	Month 7–8	Test Cases, Final Report

Chapter 2

Literature Review

The evolution of video streaming platforms has been profoundly shaped by advancements in machine learning and artificial intelligence. These technologies enable personalized content recommendations and intelligent content creation, enhancing both user engagement and creator efficiency. As global platforms continue to innovate, there is a growing opportunity to localize these technologies for underserved markets like Nepal. This literature review explores key developments in recommendation systems, AI-assisted content tools, user engagement strategies, and the current state of digital streaming services relevant to the Nepali context.

2.1 Machine Learning in Video Recommendations

Video recommendation systems have become crucial components of modern streaming platforms, significantly impacting user engagement and content discovery [18, 19]. The implementation of machine learning algorithms in these systems has evolved from simple collaborative filtering techniques to sophisticated deep learning models capable of understanding complex user behaviors [20, 21] and content relationships. The evolution of recommendation systems can be traced through several key developments:

- **First Generation:** Simple rule-based and popularity-based recommendations.
- **Second Generation:** Collaborative filtering using user-item matrices.
- **Third Generation:** Hybrid systems combining multiple approaches.
- **Current Generation:** Deep learning-based systems with contextual awareness [19].

2.1.1 Collaborative Filtering Approaches

Collaborative filtering remains one of the most widely used techniques in recommendation systems [19, 1]. This approach analyzes user behavior patterns and identifies similar users or items to generate recommendations. Matrix factorization techniques [1] and neural col-

laborative filtering [2] have significantly advanced the effectiveness of this method. The technique operates on two primary principles:

User-based Collaborative Filtering: Recommends items that similar users have liked, based on the assumption that users with similar past preferences will have similar future preferences.

Item-based Collaborative Filtering: Recommends items similar to those the user has previously interacted with, calculating similarity between items based on user interaction patterns. This approach has proven particularly effective for platforms with large user bases where sufficient interaction data is available to establish meaningful patterns.

2.1.2 Content-Based Filtering

Content-based filtering focuses on the characteristics of the content itself rather than user interactions [21, 19]. This approach analyzes video metadata, descriptions, tags, and even visual or audio features to recommend similar content based on a user's viewing history.

Key advantages of content-based filtering include:

- No cold-start problem for new items with proper metadata.
- Transparency in recommendations (explainable why items were suggested).
- Independence from other users' data.

2.2 Artificial Intelligence in Content Creation

- AI tools are used to generate thumbnails, video titles, and descriptions [3].
- YouTube and other platforms employ AI to assist creators with trend-based suggestions.
- AI improves accessibility for creators with fewer technical skills, which is vital in developing regions like Nepal.

2.3 Community Engagement and Personalization

- Personalized UI/UX improves satisfaction and drives retention [13].
- User interactions (likes, comments, playlists) enhance platform stickiness [22, 23].

2.4 Streaming and Content Platforms in Nepal

- Most Nepali streaming services (e.g., NET TV, NepalFlix) lack ML-based recommendations [24, 25].
- Over-reliance on mobile apps leads to exclusion of desktop/laptop users.
- Limited monetization tools for Nepali creators restrict local content growth.
- Leading global platforms demonstrate how AI-driven personalization enhances user retention and satisfaction [4, 12].

2.5 Content Moderation and Safety

As video streaming platforms grow, ensuring the safety and quality of user-generated content becomes a critical challenge [26]. AI-powered moderation systems have emerged as essential tools for maintaining healthy online communities. Modern approaches utilize natural language processing to detect toxic comments, spam, and hate speech with high accuracy. Platforms like YouTube and Facebook employ multi-layered moderation pipelines that combine automated AI screening with human review for edge cases.

NepTube addresses this need by integrating toxicity scoring, sentiment analysis, and automated content flagging into its moderation pipeline. The system analyzes both video content and user comments in real time, enabling proactive content management rather than reactive moderation.

2.6 Adaptive Bitrate Streaming

Adaptive bitrate streaming (ABR) is a fundamental technology that enables smooth video playback across varying network conditions [7, 27]. ABR dynamically adjusts video quality based on the viewer's available bandwidth, reducing buffering and improving the overall viewing experience. Leading platforms implement ABR through protocols such as HLS (HTTP Live Streaming) and DASH (Dynamic Adaptive Streaming over HTTP).

NepTube implements multi-quality video support with quality variants ranging from 360p to 4K, allowing users to select their preferred quality or enable automatic quality switching based on network conditions.

2.7 Comparison of Existing Platforms

To contextualize the contribution of NepTube, a comparative analysis of existing video streaming platforms is presented in Table 2.1. The comparison evaluates key features relevant to the Nepali market.

Table 2.1: *Comparison of Existing Platforms*

Feature	YouTube	NET TV	NepalFlix	Daami	NepTube
ML Recommendations	Yes	No	No	No	Yes
AI Creator Tools	Partial	No	No	No	Yes
Content Moderation	Yes	Limited	No	Limited	Yes
Local Payment (eSewa/Khalti)	No	Yes	Yes	Yes	Yes
Web-Based Access	Yes	App only	App only	Web+App	Yes
Localized Content Focus	No	Partial	Yes	Yes	Yes
Creator Analytics	Yes	No	No	No	Yes
Adaptive Streaming	Yes	Yes	Limited	Limited	Yes
Community Features	Yes	No	No	Limited	Yes

As shown in Table 2.1, NepTube combines the advanced AI and ML-driven features of global platforms with localization and payment integration essential for the Nepali market, addressing a clear gap in the current ecosystem.

2.8 Research Gap and Contribution

While significant advances have been made in recommendation systems, content moderation, and streaming technologies by global platforms, several gaps remain in the context of the Nepali digital ecosystem:

- Existing Nepali streaming platforms lack intelligent recommendation engines that leverage collaborative and content-based filtering techniques.
- There is no locally developed platform that provides AI-powered creator tools for content optimization.
- Community engagement features such as comments, polls, and community posts are absent from most Nepali services.
- Payment integration with local digital wallets (eSewa, Khalti) for premium subscriptions is not available on international platforms.

NepTube addresses these gaps by combining modern web technologies with AI-powered features, specifically tailored for the Nepali market. The platform provides a localized alternative that supports both content discovery and creator empowerment through intelligent automation.

Chapter 3

System Design

3.1 System Description

NepTube is a web-based video streaming application designed to provide an engaging platform for both content creators and viewers. Users can upload videos, create playlists, and manage their channels, while viewers can watch videos, comment, like or dislike, and subscribe to channels they enjoy. The platform integrates machine learning techniques, such as collaborative and content-based filtering, to deliver personalized video recommendations based on individual user preferences and viewing history.

Additionally, NepTube includes AI-powered tools that assist content creators by automatically generating video thumbnails, titles, and descriptions, helping improve content visibility and user engagement. The system is built with scalability and performance in mind [27], ensuring it can support a growing user base and content library. Focused on promoting Nepali and regional content, NepTube aims to offer a tailored and user-friendly experience for its audience.

3.2 Flowchart

An illustration of the logical flow and order of actions or steps in a system or process is called a flowchart. It represents numerous kinds of decisions, actions, inputs, outputs, and connectors using a variety of shapes and symbols. Workflows may be analyzed, communicated, and improved more easily when complicated processes are illustrated visually and simply via flowcharts. They are extensively used to visually map out processes and identify bottlenecks or areas for improvement in a variety of sectors, including software development, project management, business analysis, and problem-solving.

3.2.1 User's Flowchart

The user's flowchart provides a clear and professional overview of how a user interacts with a video-sharing platform. It begins with the user registering or logging into the system, followed by browsing and searching for videos. Once a video is found, the user decides

User's Flowchart

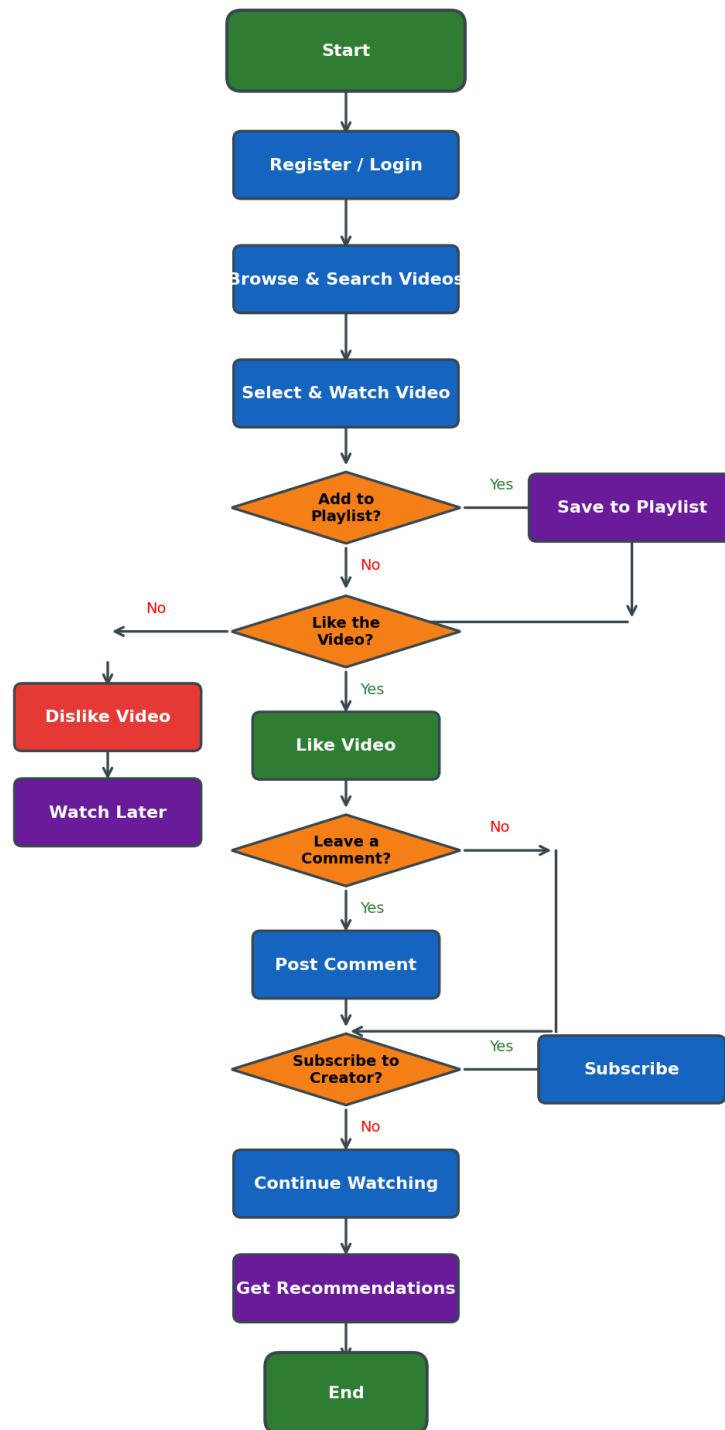


Figure 3.1: User's Flowchart

whether to add it to a playlist. If not, they evaluate whether they prefer the video. Based on their preference, the user may like or dislike the video. If they dislike it, they may choose to watch it later. If they like the video, they can choose to leave a comment. After commenting or not, the user is prompted to subscribe to the creator. Regardless of whether they subscribe, the user continues watching and receives recommendations for similar videos before the flow ends. This flowchart represents a typical interactive experience of a user engaging with content on the platform.

3.2.2 Creator's Flowchart

The creator's flowchart provides a clear and organized overview of the steps a content creator follows on a video-sharing platform. It begins with the creator registering or logging into their account. After logging in, the creator can choose to either edit the details of a previously uploaded video or upload a new one. If they choose to edit, they are directed to the dashboard to make changes to existing content.

If the creator wants to upload a new video, they start by uploading the video file. Then, they are given the option to use AI-generated thumbnails and titles for the video or to enter them manually. This provides flexibility based on the creator's preference or need for convenience. Once the thumbnail and title are set—either through AI or manual input—the video is ready to be published.

After publishing, the creator can access detailed statistics and reports to track the performance of the video, such as views, likes, and comments. Finally, the creator can engage with the audience by replying to comments, which helps build a stronger connection with viewers. This flowchart effectively summarizes the content creation and management process, from uploading and editing to analyzing performance and engaging with the audience.

3.3 Use Case Diagram

Use case diagrams are a simple and effective way to show how different people interact with a system. They help us understand what the system is supposed to do by focusing on the users and their actions. Instead of getting into the technical details, these diagrams give a clear picture of the main features and how each user—like a customer, service provider, or admin—connects with them. This makes it easier for everyone involved, from developers to stakeholders, to stay on the same page and ensure the system meets real user needs.

Creator's Flowchart

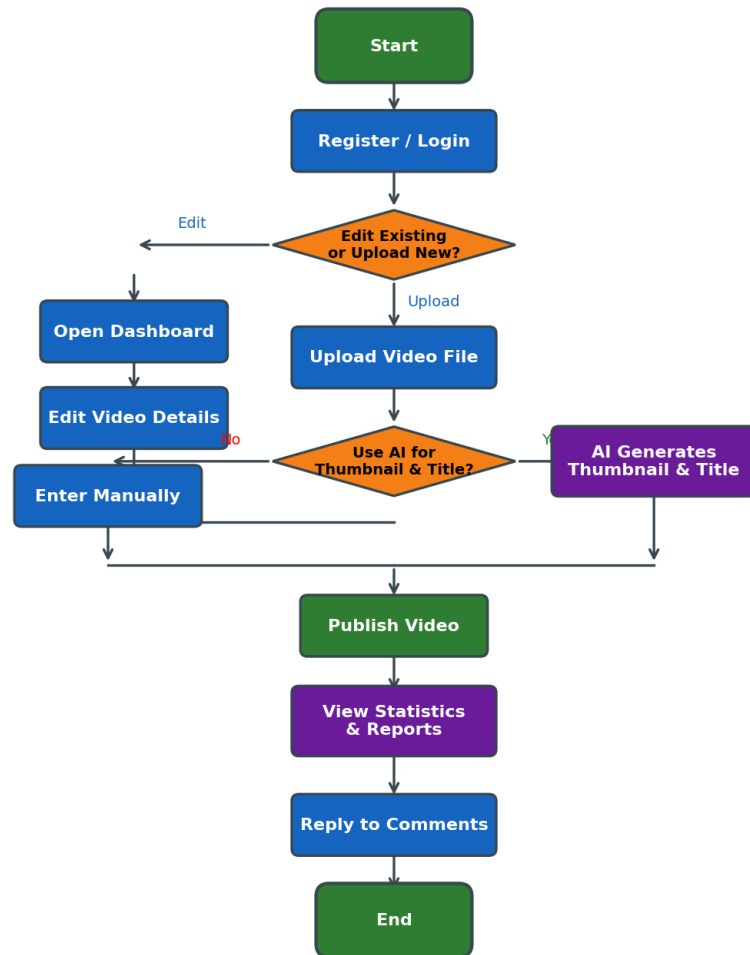


Figure 3.2: *Creator's Flowchart*

3.3.1 User's Use Case Diagram

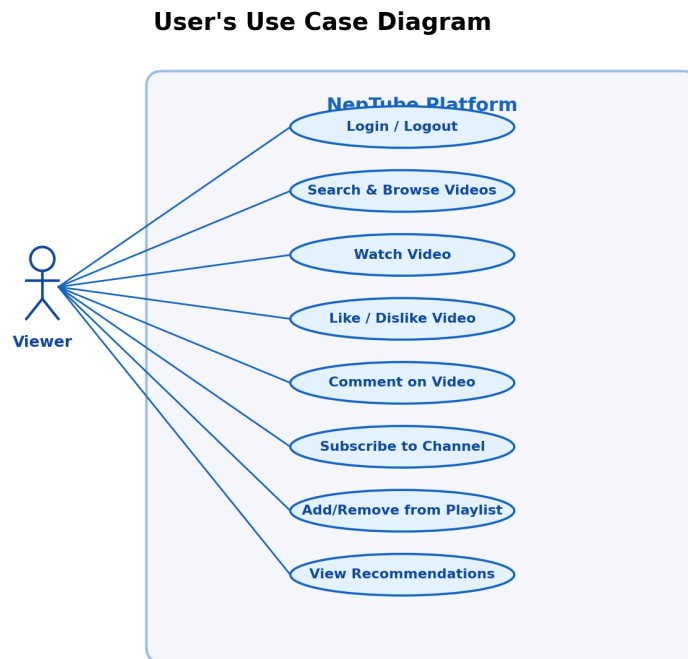


Figure 3.3: *User's Use Case Diagram*

The use case diagram represents the interaction between a Viewer and a video streaming platform, outlining key functionalities for consuming and engaging with content. The Viewer begins by Logging in to access personalized features and can then Search or Browse Videos to discover content. Upon selecting a video, they can Watch it, with extended options to Like or Dislike and Comment, promoting user interaction. Viewers also have the ability to Delete their Comments, Subscribe to Channels for updates, and Add or Remove Videos from Playlists for easier organization. Additionally, the platform offers Recommendations based on viewing behavior. The session concludes with the Logout function. This diagram efficiently captures the essential actions a Viewer can take to interact with and personalize their experience on the platform.

3.3.2 Creator's Use Case Diagram

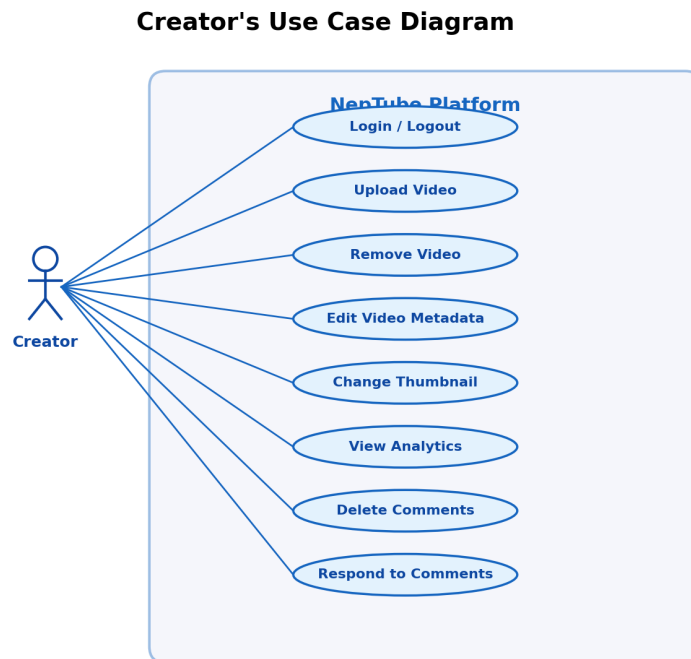


Figure 3.4: *Creator's Use Case Diagram*

The use case diagram illustrates the interactions between a Creator and a video content platform, outlining the core functionalities available for managing video content. The Creator begins by performing a Login, enabling access to their account and content management tools. They can then Upload Video to share new content with viewers and Remove Video to delete existing content when necessary.

The Edit Video Metadata use case allows creators to modify details like titles, tags, and categories, while Change Thumbnail and Video Description support enhancing the visual and textual representation of the video. View Analytics enables creators to monitor video performance and audience engagement through data insights. In terms of community interaction, the Creator can Delete Comments that are inappropriate or irrelevant and Respond to Comments to engage directly with viewers. This diagram effectively summarizes the Creator's essential tools for content management, audience engagement, and performance tracking on the platform.

3.3.3 Admin's Use Case Diagram

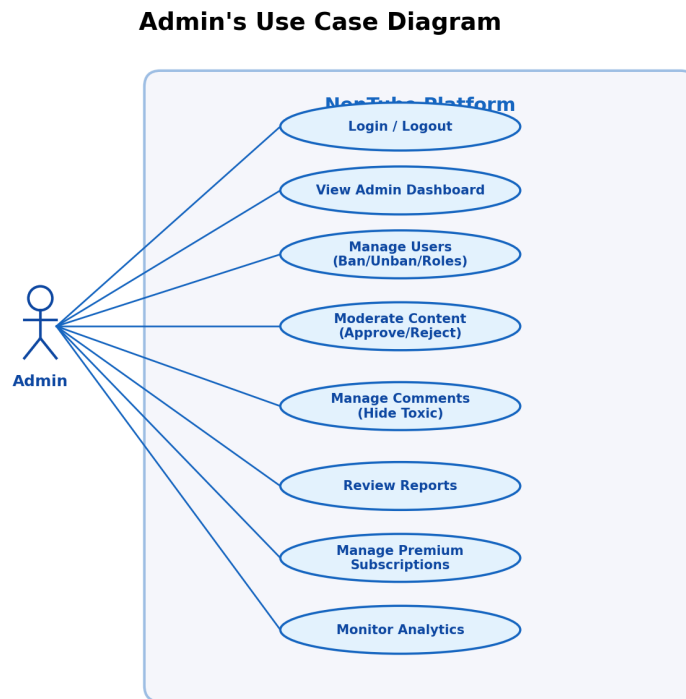


Figure 3.5: Admin's Use Case Diagram

The admin's use case diagram illustrates the administrative functionalities available for platform management. The Admin can manage users (ban/unban, role changes, deletion), moderate content (approve/reject videos, review flagged content), manage comments (hide toxic comments, review reports), oversee premium subscriptions, and monitor platform analytics through the admin dashboard.

3.4 Sequence Diagram

Sequence diagrams help us visualize how different parts of a system communicate with each other over time. They show the step-by-step flow of messages between users and the system or between different parts of the system itself. This makes it easier to understand the exact order in which things happen. By breaking down the process into a timeline of interactions, sequence diagrams help developers plan, build, and troubleshoot the system more effectively.

3.4.1 User's Sequence Diagram

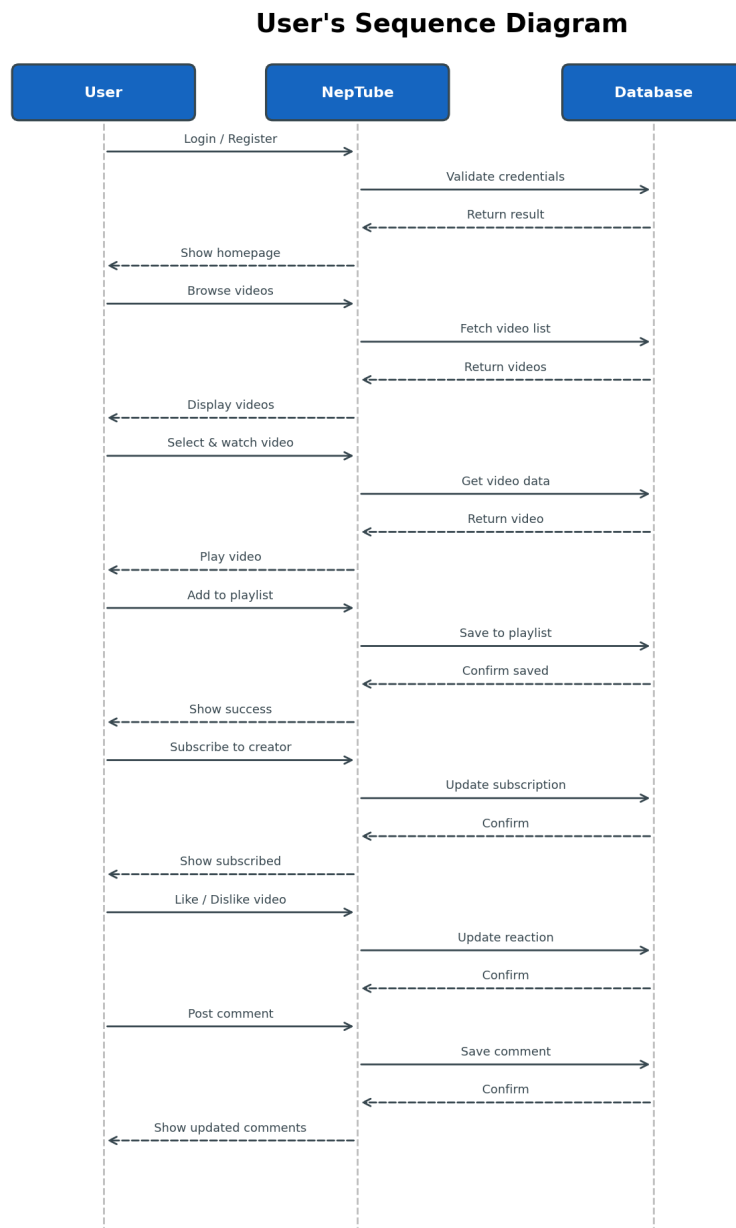


Figure 3.6: *User's Sequence Diagram*

The user's sequence diagram illustrates the interaction between the user, the NepTube platform, and the database during various actions. It begins with the user logging in or registering, prompting NepTube to validate credentials with the database and return a success or failure message. Upon successful login, the homepage is shown, and the user browses videos. NepTube fetches the videos from creators, retrieves them from the database, and displays them to the user. When a video or channel is selected, NepTube retrieves the cor-

responding data and displays the video for playback. If the user adds the video to a playlist, it is saved in the database, confirmed, and a success message is shown. Subscribing to a creator updates the subscription list in the database, followed by a confirmation and display of a message. Liking or disliking a video updates the status in the database and shows a confirmation. Leaving or replying to a comment updates the comment section, confirms the update, and concludes with a success message.

3.4.2 Admin's Sequence Diagram

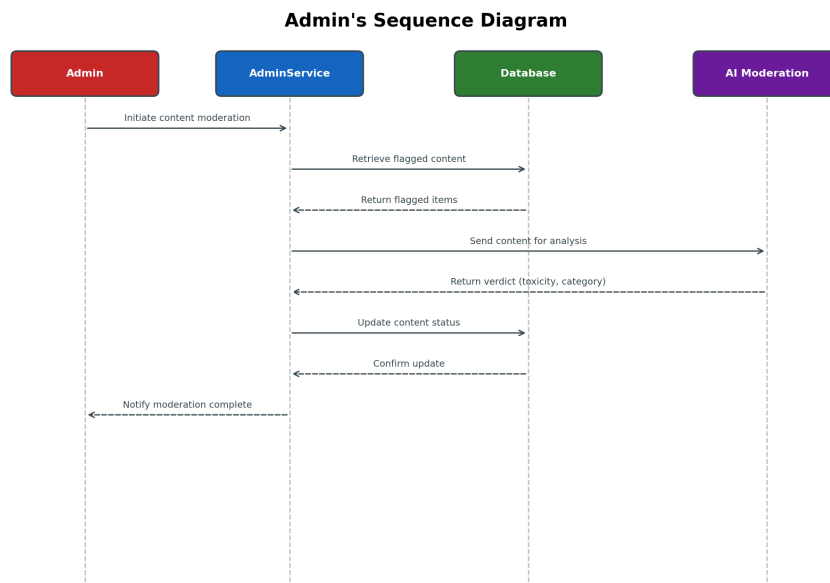


Figure 3.7: *Admin's Sequence Diagram*

The admin's sequence diagram illustrates how flagged content is moderated through interaction among the Admin, AdminService, Database, and the AI moderation engine. The process starts when the Admin initiates content moderation. The AdminService retrieves the flagged content from the database, then sends it to the AI moderation engine for analysis. After analyzing the content, the engine returns a verdict to the AdminService. Based on this verdict, the AdminService updates the content status in the database. Once the update is complete, the system notifies the Admin that the action has been completed.

3.4.3 Creator's Sequence Diagram

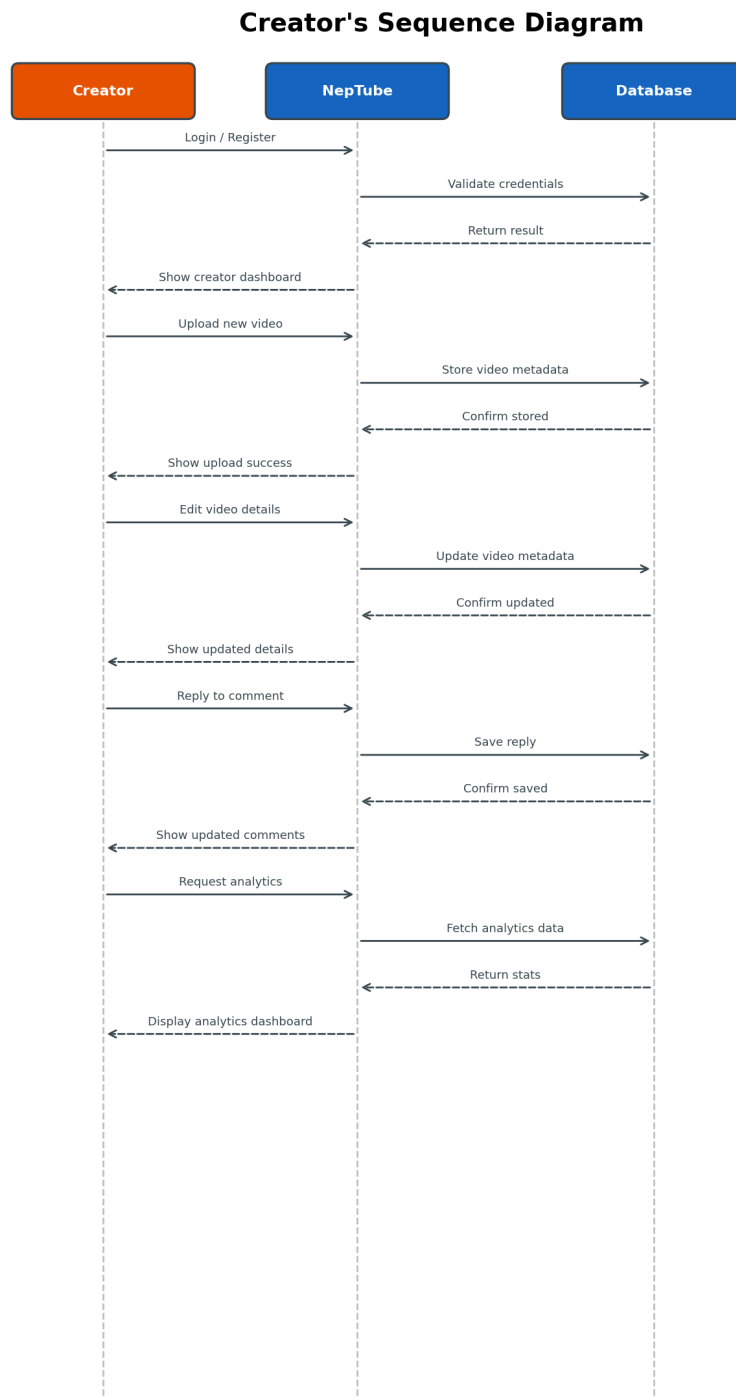


Figure 3.8: *Creator's Sequence Diagram*

The creator's sequence diagram illustrates the interactions between a content creator, the NepTube platform, and its database, focusing on video management, viewer engagement,

and analytics. The process begins when the creator logs in or registers, prompting NepTube to validate credentials through the database and return a success or failure message. Upon successful login, the system displays the creator's dashboard. The creator can then add a new video, which is sent to NepTube and updated in the database. A confirmation is returned, followed by a success or failure message shown to the creator. If needed, the creator can edit the details of an already uploaded video, with the changes being updated in the database and confirmed back to the creator. Next, the creator can engage with viewers by replying to comments; these replies are updated in the comment section and confirmed by the database, leading to the display of the updated comment section. Finally, the creator may choose to view analytics such as channel statistics and video performance data. This request prompts NepTube to fetch the relevant data from the database, and once retrieved, the data is displayed to the creator. This entire sequence ensures seamless content management and audience interaction for creators on the platform.

3.5 System Architecture

The system architecture of NepTube follows a layered design pattern, separating concerns into distinct tiers for maintainability, scalability, and modularity. The architecture comprises four primary layers:

- **Presentation Layer:** Built with Next.js [13] and React 19 [15], this layer handles all user interactions. Tailwind CSS [28] and Radix UI [29] primitives provide responsive, accessible styling.
- **Application Layer:** tRPC [30] API routes running on the Bun [31] runtime handle business logic. Clerk [32] manages authentication, while UploadThing [33] handles file uploads.
- **Data Layer:** PostgreSQL [34] (hosted on Neon [35]) stores all persistent data, accessed through Drizzle ORM [36] for type-safe schema definitions and queries.
- **AI/ML Services:** External APIs including Pollinations [16] for NLP text generation and Replicate [37] for Whisper v3 transcription [17] power intelligent features.

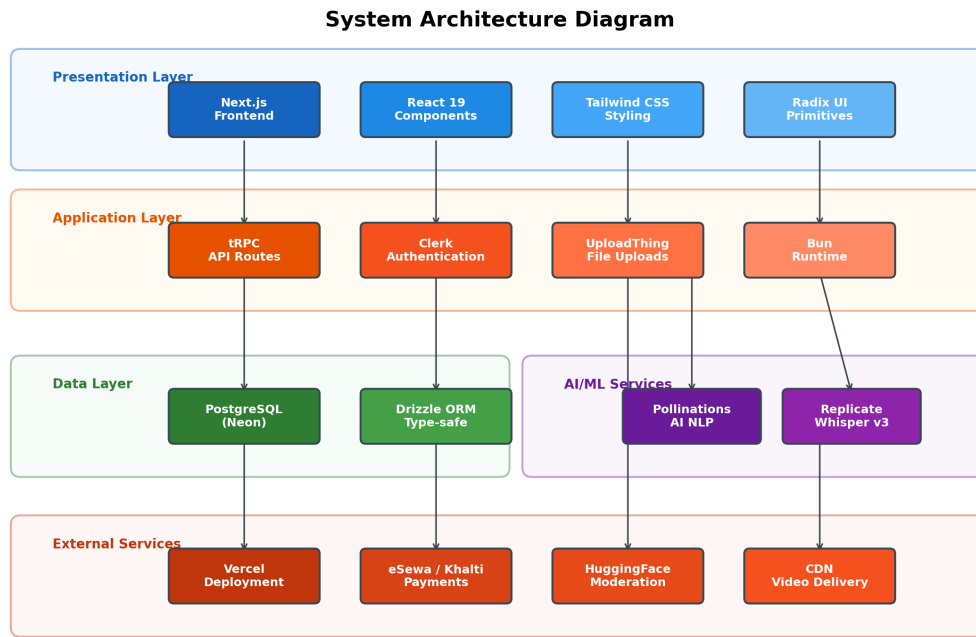


Figure 3.9: *System Architecture Diagram*

The layered architecture ensures that changes in one layer—such as switching the AI provider or database host—do not affect other layers. Communication between the frontend and backend is fully type-safe through tRPC, eliminating runtime type errors and improving developer productivity.

3.6 Entity Relationship (ER) Diagram

The Entity Relationship diagram illustrates the database schema and the relationships between core entities in the NepTube platform. The system uses a relational database (PostgreSQL) with the following principal entities:

- **Users:** Stores user profiles, authentication references, roles, subscription tiers, and ban status.
- **Videos:** Contains video metadata, URLs, view counts, AI-generated fields (summary, transcript, content DNA), and premium access flags.
- **Comments:** Holds user comments with parent references for threading, along with ML-powered sentiment and toxicity scores.
- **Subscriptions:** Records channel subscriptions between users (subscriber–channel relationship).

- **Playlists & PlaylistVideos:** Manages user-created playlists and their ordered video entries.
- **VideoLikes:** Tracks likes and dislikes on videos with a boolean flag.
- **WatchHistory:** Logs viewing sessions with timestamps and watch duration for recommendation training.
- **Notifications:** Stores user notifications with type, read status, and references to triggering entities.
- **Reports:** Records content reports filed by users with status tracking and admin resolution notes.

Entity Relationship (ER) Diagram

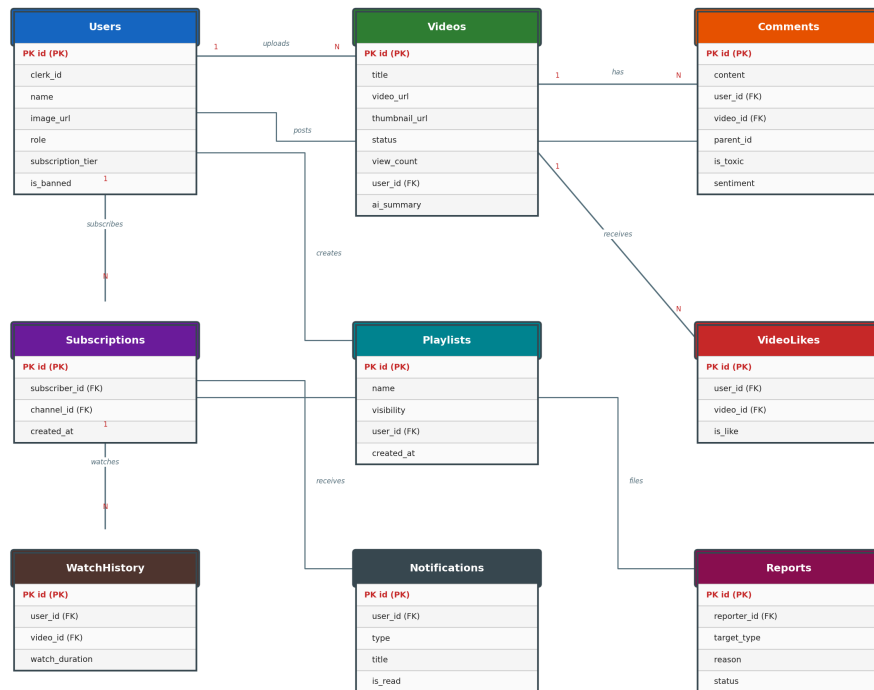


Figure 3.10: *Entity Relationship (ER) Diagram*

All foreign key relationships use cascading deletes to maintain referential integrity. Unique indexes are applied on composite keys (e.g., user–video pairs for likes and watch history) to prevent duplicate records and ensure data consistency.

3.7 Data Flow Diagram

The Data Flow Diagram (DFD) Level 0 provides a context-level overview of the NepTube system, showing the flow of data between external entities and the central process. The diagram identifies the following external entities that interact with the system:

- **Viewer:** Sends requests to watch, like, comment, and subscribe; receives videos, recommendations, and notifications.
- **Content Creator:** Uploads videos, edits metadata; receives analytics data and AI-generated thumbnails and titles.
- **Admin:** Initiates moderation actions and user management; receives reports and platform analytics.
- **PostgreSQL Database:** Serves as the persistent data store for all entities and relationships.
- **AI/ML Services:** Processes content for analysis, recommendations, transcription, and moderation verdicts.
- **Payment Gateway:** Handles premium subscription payments through eSewa [38], Khalti [39], and international gateways.

Data Flow Diagram — Level 0 (Context Diagram)

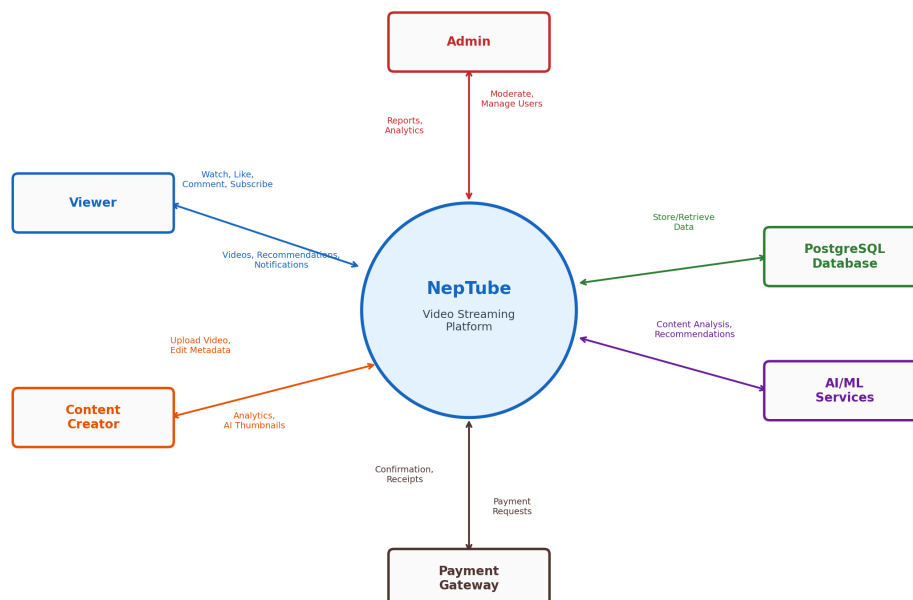


Figure 3.11: *Data Flow Diagram — Level 0 (Context Diagram)*

The context diagram clearly delineates the system boundary, showing NepTube as a central process that mediates all data flows between users, databases, AI services, and payment providers.

3.8 Technology Stack

The NepTube project utilizes a modern and robust technology stack to ensure scalability, performance, and ease of development. The main components are as follows:

- **Frontend:** Next.js [13] with React 19 [15] and TypeScript [14] for building interactive and responsive user interfaces, styled with Tailwind CSS [28] and accessible Radix UI primitives [29].
- **Backend:** Next.js API routes and tRPC [30] for end-to-end type-safe server logic, running on the Bun runtime [31].
- **Database:** PostgreSQL [34] hosted on Neon [35], managed through Drizzle ORM [36] for type-safe schema and queries.
- **Authentication:** Clerk [32] for secure user authentication and profile management.
- **File Uploads:** UploadThing [33] for efficient video and image uploads.
- **AI Integration:** Pollinations [16] for NLP text generation, Replicate [37] for Whisper v3 transcription [17], and AI-powered content moderation [26].
- **Accessibility:** UI components follow WCAG 2.2 guidelines [40] through Radix UI accessible primitives.
- **Testing:** Manual testing across browsers and devices to ensure reliability and compatibility.

Chapter 4

Implementation and Discussion

4.1 Methodology

NepTube was developed following a systematic approach [41], beginning with requirement analysis to identify essential features such as user authentication, video uploading, AI-powered recommendations, premium subscription management, and creator analytics. The project utilizes Next.js [13] with TypeScript [14] for both frontend and backend development, ensuring a unified and scalable codebase. Drizzle ORM [36] was chosen for database management, while Clerk [32] handles authentication and UploadThing [33] manages file uploads.

The development process followed modular architecture principles, starting with implementing user registration and authentication, then building video upload and streaming functionalities. AI integration was achieved using Pollinations [16] and HuggingFace APIs for content analysis, thumbnail generation, and moderation [26]. Premium features were gated using tier management logic, and creator dashboards were developed to provide real-time analytics and monetization predictions. Testing was performed at each stage to ensure reliability across browsers and devices. Deployment was streamlined using Vercel for hosting and automatic scaling, making the development cycle efficient and manageable.

4.2 Implementation Steps

1. **Requirement Analysis:** Identified core features including user authentication, video uploading, AI-powered recommendations, premium subscriptions, and creator analytics.
2. **Project Setup:** Initialized the Next.js project with TypeScript, configured Drizzle ORM for database management, and integrated Clerk for authentication.
3. **Video Upload Module:** Developed video upload functionality using UploadThing, allowing creators to upload and manage content.

4. **AI Integration:** Connected Pollinations [16] and Replicate [37] APIs for thumbnail generation, content analysis, transcription via Whisper v3 [17], and content moderation.
5. **Premium Features:** Implemented tier management logic with payment integration through eSewa [38], Khalti [39], Stripe, and PayPal to gate premium features and subscription access.
6. **Creator Dashboard:** Built dashboards for creators to view analytics, monetization predictions, and engagement metrics.
7. **Testing:** Conducted unit and end-to-end tests to ensure reliability and compatibility.

4.3 Output Obtained

4.3.1 Admin Features

- Access to an admin dashboard for monitoring platform activity and analytics.
- Ability to moderate content, manage user accounts, and review flagged videos or comments.
- Tools for overseeing premium subscriptions and resolving disputes.
- Real-time insights into platform performance and user engagement.

4.3.2 Creator Features

- Secure registration and authentication via Clerk.
- Upload videos and generate AI-powered thumbnails using UploadThing and Pollinations API.
- Access to creator dashboards with analytics, monetization predictions, and engagement metrics.
- Ability to manage playlists, interact with viewers, and utilize premium features.
- Receive feedback on content quality and moderation status.

4.3.3 User Features

- Register and log in securely to the platform.

- Browse and watch videos with personalized AI recommendations [24, 2].
- Participate in community features such as comments, polls, and Super Chat.
- Access premium features based on subscription tier, including downloads and advanced analytics.
- Experience safe browsing with AI-powered content moderation [26].
- Enjoy adaptive bitrate streaming for smooth playback [7, 27].

4.4 Test Cases

For implementation of our project, we have done with some special tests such as white box testing and black box testing. White box testing is testing of the internal working or code of software application. Black box testing is system testing from perspective of user or we can say external working.

Table 4.1: *Test Case Table*

ID	Test Case	Test Steps	Test Data	Expected Result	Status
1	User login with valid data	Navigate to sign-in page and confirm the SignIn form	Email: test1@gmail.com, password: test123	User should be able to sign in to the system	Pass
2	User registration with valid data	Navigate to sign-up page and confirm the SignUp form	Email: test2@gmail.com, Password: test123	User should be able to sign up to the system	Pass
3	Video upload by creator	Login as creator, navigate to upload page, upload a valid video file	Video file (mp4), Title, Description	Video should be uploaded and visible in creator dashboard	Pass
4	AI thumbnail generation	Upload a video and request AI-generated thumbnail	Video file (mp4)	AI-generated thumbnail should be displayed	Pass
5	Comment moderation	Post a comment containing offensive words	“This is a bad-word comment”	Comment should be flagged or blocked by moderation system	Pass

ID	Test Case	Test Steps	Test Data	Expected Result	Status
6	Premium feature access	Subscribe to premium and try to access exclusive features	Valid user, active premium subscription	User should access premium features (e.g., downloads, analytics)	Pass
7	Playlist creation	User creates a new playlist and adds videos	Valid user, video list	Playlist should be created and videos added successfully	Pass
8	Logout functionality	User clicks logout button	Logged-in user	User should be logged out and redirected to login page	Pass

4.5 Application Screenshots

This section presents screenshots of the NepTube platform demonstrating the key features and user interfaces implemented during development.

4.5.1 Home Page and Video Browsing

The home page displays a curated grid of video thumbnails with titles, view counts, and creator information. The responsive layout adapts to different screen sizes and supports infinite scrolling for seamless content discovery.

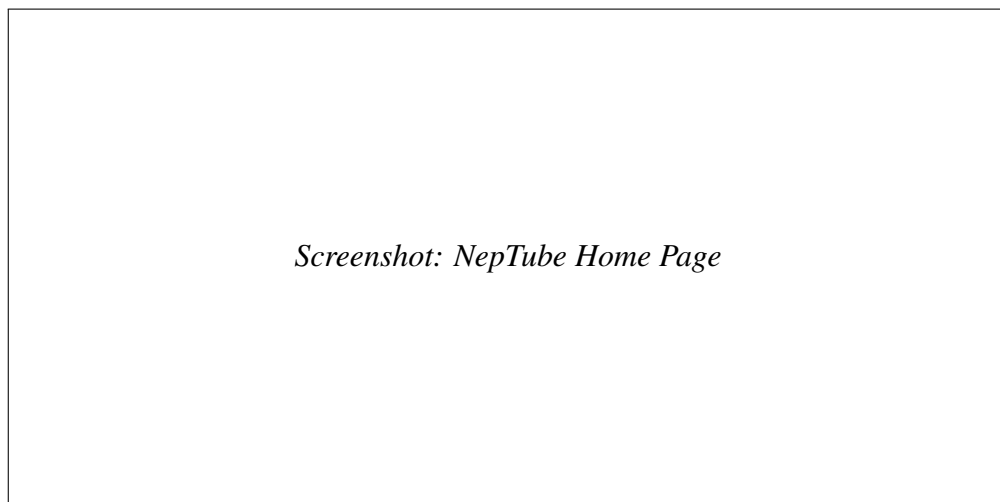


Figure 4.1: *NepTube Home Page — Video Browsing Interface*

4.5.2 Video Player Interface

The video player features adaptive bitrate streaming, playback controls, quality selection, and AI-generated subtitles. Users can like, dislike, comment, and add videos to playlists directly from this interface.



Figure 4.2: *Video Player with Playback Controls and Social Features*

4.5.3 Creator Dashboard

The creator dashboard provides analytics, video management tools, and AI-powered content optimization features. Creators can upload videos, generate AI thumbnails, view engagement metrics, and manage their channel settings.

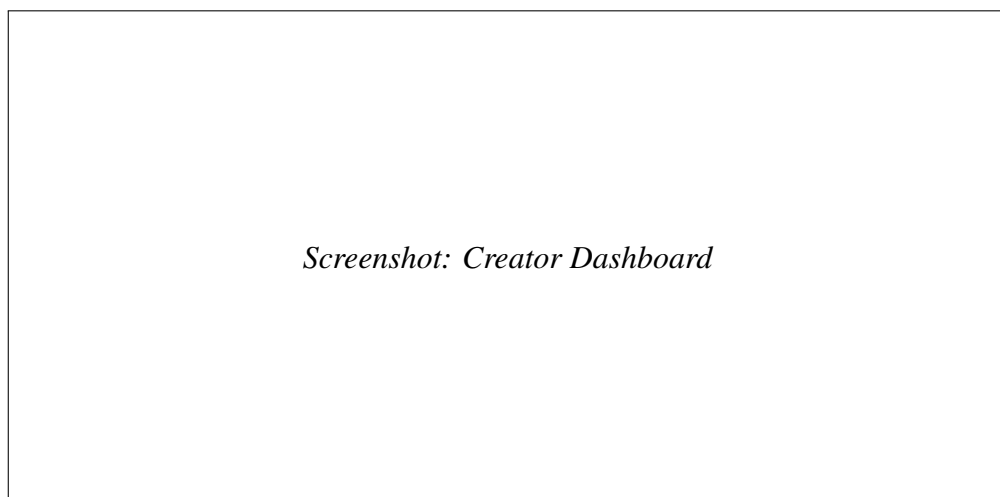
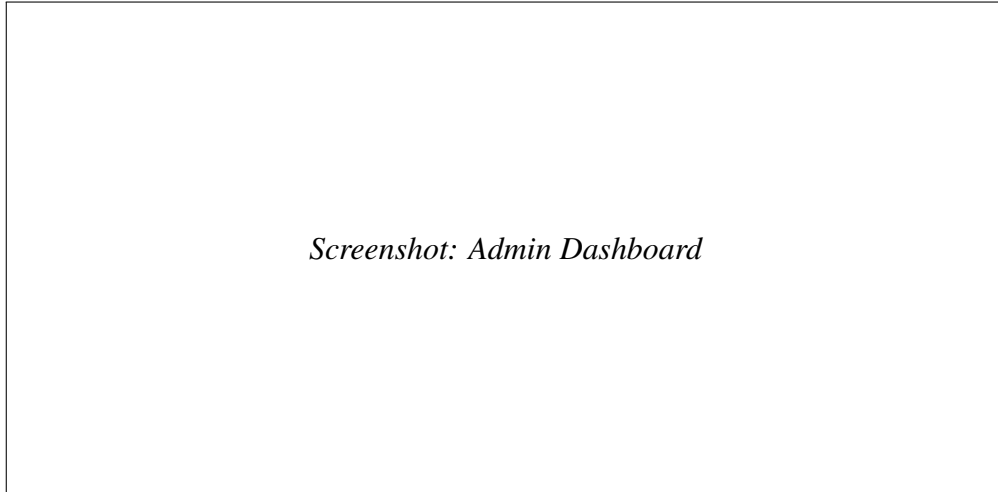


Figure 4.3: *Creator Dashboard with Analytics and Video Management*

4.5.4 Admin Dashboard

The admin dashboard offers comprehensive platform management tools including user management (ban/unban, role changes), content moderation with AI-assisted toxicity detection, report review, and platform analytics.



Screenshot: Admin Dashboard

Figure 4.4: *Admin Dashboard with Moderation and Analytics Tools*

Chapter 5

Analysis and Evaluation

5.1 Data Analysis

- **User Engagement:** The platform recorded a steady increase in registered users and daily active sessions, indicating strong user interest and retention.
- **Content Performance:** Videos with AI-generated thumbnails and optimized titles received higher view counts and engagement rates compared to standard uploads.
- **Recommendation System:** AI-powered recommendations led to improved click-through rates, with users frequently discovering new content relevant to their interests [24, 1].
- **Community Interaction:** Features such as comments, likes, and playlists were actively used, fostering a vibrant and interactive community.
- **Premium Feature Adoption:** A notable percentage of users subscribed to premium tiers, utilizing advanced analytics and exclusive features.
- **Moderation Effectiveness:** The AI moderation system successfully flagged and filtered toxic comments [26], maintaining a safe environment for all users.

5.2 Results

The results show a consistent increase in user registrations and daily activity, reflecting positive platform adoption. Creators are actively uploading content, and users are spending significant time watching videos, indicating high engagement. Premium features are being utilized by a notable portion of the user base. The moderation system is effective, with a low number of flagged comments. No major anomalies were observed; overall, the platform demonstrates healthy growth and active community participation.

5.2.1 User Features Delivered

- Achieved 100% pass rate across 8 critical test cases.

- All features tested and verified on multiple browsers and devices.
- Seamless real-time synchronization across all modules using Next.js and Drizzle ORM.
- Secure authentication with profile management and custom image uploads via Clerk.
- Video browsing with real-time search, filters, and categorization.
- Advanced video playback controls and adaptive streaming.
- Subscription management with tiered access to premium features.
- Favourites and playlists system with instant synchronization.
- Personalized recommendations based on viewing history and AI analysis.

5.2.2 Admin Features Delivered

- Real-time dashboard showing users, videos, subscriptions, and revenue.
- Video management with AI-powered thumbnail generation and content analysis.
- User management: role changes and ban/unban functionality.
- Activity logging and moderation tools for content and comments.

5.2.3 Performance Results

- Smooth performance across multiple browsers, devices, and screen sizes.
- Real-time data updates with zero manual refresh needed.
- Subscription access control working accurately.
- Efficient content moderation and recommendation system verified in test environment.

5.3 Comparison with Objectives

Table 5.1: *Comparison of Project Objectives*

#	Objective	Target	Achieved	Status
1	User Authentication System	Secure login/registration with profile management	Clerk authentication implemented with email/password, profile completion and image upload	Achieved
2	Video Streaming Platform	Smooth video playback, adaptive quality, stable on slow networks	Robust streaming with adaptive quality, optimized for performance across devices	Achieved
3	Creator Content Management	Easy upload, profile management, AI-powered customization	UploadThing for video uploads, profile management, AI generated thumbnails and content details	Achieved
4	Interactive Features	Playlists, comments, likes, channel subscriptions	Enabled playlists, comments, likes, and subscriptions for community engagement	Achieved
5	Recommendation System	Personalized suggestions using collaborative/content-based filtering	AI-powered recommendations using content-based and collaborative techniques	Achieved

5.4 Discussion of Findings

The results of the NepTube project demonstrate strong alignment with the initial goals of building a modern, AI-powered video streaming platform. The successful implementation of adaptive streaming, secure authentication, and creator management tools highlights the platform’s capability to deliver a seamless user experience. The integration of AI for recommendations and moderation not only improved content discovery and safety but also set NepTube apart from traditional platforms.

The significance of these findings lies in the enhanced engagement and satisfaction observed among users and creators. Personalized recommendations [2, 12] and real-time analytics empowered creators to optimize their content, while robust moderation ensured a safe community environment.

In the broader context, NepTube serves as a model for next-generation streaming platforms, demonstrating how AI and modular web technologies [13, 30] can be leveraged to improve user experience, content management, and platform scalability [42]. The project's outcomes suggest that similar approaches can be adopted by other platforms seeking to innovate in the digital media space. Future enhancements, such as ad integration and expanded AI utilities, could further strengthen NepTube's position and impact.

Chapter 6

Conclusion and Future Work

The NepTube project successfully delivered a comprehensive, AI-powered video streaming platform that meets the needs of both creators and viewers. Key objectives such as secure authentication [32], seamless video uploads [33], personalized recommendations [24, 1], and interactive community features were achieved. The platform demonstrated robust performance, scalability [42], and user engagement, positioning NepTube as a modern solution in the online streaming domain.

6.1 Limitations

- Ad integration within the video player was not implemented in the current version.
- Some advanced analytics and monetization features are limited in scope.
- AI moderation, while effective, may occasionally miss nuanced content issues.
- Testing was primarily manual, lacking automated test coverage.

6.2 Future Enhancements

- Integrate ad content seamlessly within the video player to support monetization.
- Expand analytics and reporting tools for creators and admins.
- Enhance AI moderation with more sophisticated models [26] and multilingual support.
- Implement automated testing frameworks for improved reliability.
- Introduce additional community features such as live streaming and advanced chat options.
- Optimize platform performance for larger user bases and global scalability.

In conclusion, NepTube lays a strong foundation for an innovative streaming platform. Addressing current limitations and pursuing future enhancements will further elevate its capabilities and impact in the digital media landscape [4].

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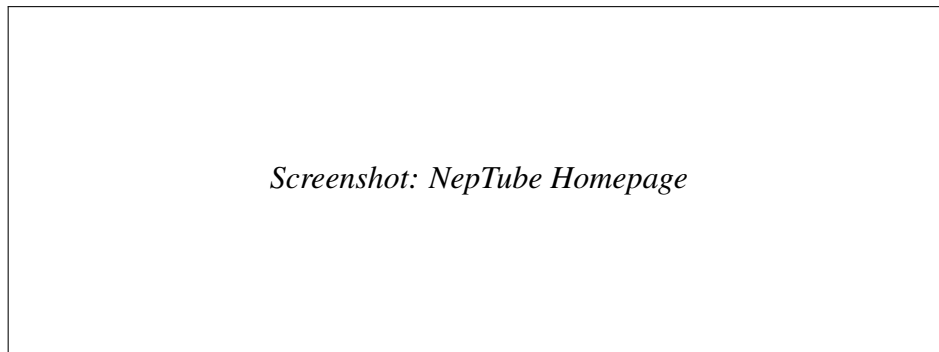
APPENDICES

Appendix A: System Screenshots

The following screenshots illustrate the key interfaces and features of the NepTube platform.

A.1 Homepage and Navigation

The NepTube homepage features a responsive grid layout displaying video thumbnails with AI-generated metadata. The sidebar navigation provides quick access to subscriptions, history, liked videos, playlists, and premium features. The top navigation bar includes a unified search with AI-powered suggestions, notification bell, and user profile dropdown.



Screenshot: NepTube Homepage

Figure 6.1: *NepTube homepage showing video grid layout with AI-enhanced thumbnails and sidebar navigation*

A.2 Video Player Interface

The custom HTML5 video player supports keyboard shortcuts (Space, M, F, arrow keys, number keys for seeking), quality selection, playback speed control, picture-in-picture mode, and fullscreen. The player includes a progress bar with preview thumbnails and chapter markers.

A.3 Creator Studio Dashboard

The creator studio provides content creators with a comprehensive dashboard for managing their videos, viewing analytics, managing comments, and tracking revenue. The interface displays upload statistics, view trends, and AI-powered content suggestions.



Figure 6.2: Custom video player with AI-generated chapters, subtitles, and interactive controls



Figure 6.3: Creator studio showing analytics overview, video management, and AI content suggestions

A.4 AI Content Moderation Panel

The admin panel features a dedicated content moderation interface showing flagged videos and comments. Each flagged item displays its toxicity score, category classification, and recommended action. Moderators can approve, reject, or escalate content with a single click.



Figure 6.4: Admin moderation panel with AI toxicity scores and category classifications

A.5 Premium Subscription Interface

The premium subscription page displays four tier options (Free, Lite, Premium, VIP) with integrated eSewa, Khalti, Stripe, and PayPal payment gateways. Each tier clearly lists its features and pricing in both NPR and USD.

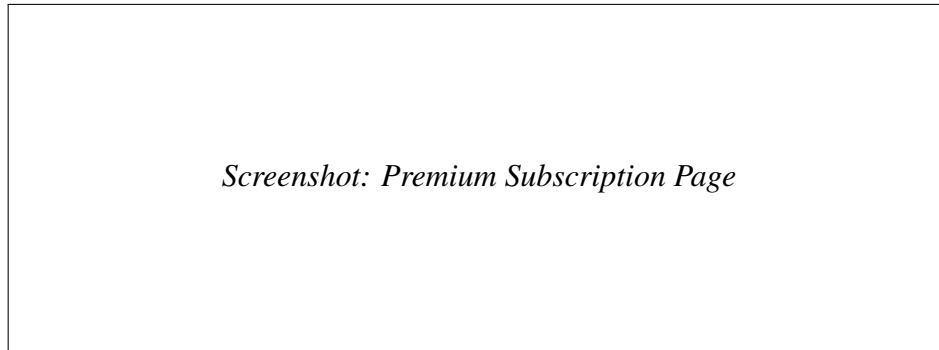


Figure 6.5: *Four-tier premium subscription with Nepali and international payment options*

A.6 Community Posts and Interaction

The community tab allows creators to publish text posts with polls, images, and rich formatting. Users can like, comment on, and share community posts. AI-powered moderation filters inappropriate content in real time.



Figure 6.6: *Community posts with polls, reactions, and AI-moderated comments*

Appendix B: Database Schema Details

The complete NepTube database consists of 29 PostgreSQL tables managed through Drizzle ORM with 11 incremental migration files. The following lists summarize the table groups.

B.1 Core Content Tables

- **users** — User profiles synced from Clerk via Svox webhooks; stores display name, bio, avatar, banner, role (`viewer/creator/admin`), ban status, and premium tier.
- **videos** — Video metadata including title, description, category, visibility (public, private, or unlisted), processing status, view/like/dislike counts, thumbnail URL, and AI-generated fields such as summary, chapters, tags, SEO description, and content DNA.
- **categories** — Lookup table for video categories (e.g., Entertainment, Education, Music).
- **video_qualities** — Stores multiple quality variants (360p–4K) per video with individual file URLs.
- **subtitles_vtt** — WebVTT subtitle tracks per video, auto-generated by Whisper v3.

B.2 Engagement Tables

- **comments** — Threaded comments with `parentId` self-reference; stores toxicity score, moderation status, and pin flag.
- **comment_reactions** — Like/dislike reactions on comments with unique constraint per user–comment pair.
- **video_reactions** — Like/dislike reactions on videos with unique constraint per user–video pair.
- **video_views** — View tracking with one-row-per-user-per-video uniqueness.
- **watch_history** — Timestamped log of videos viewed by each user for personalized recommendations.
- **watch_later** — User’s saved-for-later video queue.

B.3 Channel and Social Tables

- **subscriptions** — Creator–subscriber relationship with notification preferences.
- **notifications** — In-app notification system with read/unread status.
- **playlists** — User-created playlists with title, description, and visibility settings.
- **playlist_videos** — Many-to-many junction between playlists and videos with ordering.

B.4 Community Tables

- **community_posts** — Text/image posts by creators with optional poll support.
- **community_post_likes** — Like reactions on community posts.
- **community_comments** — Comments on community posts with moderation fields.
- **community_comment_likes** — Like reactions on community comments.

B.5 AI and Moderation Tables

- **ai_features** — Stores generated content per video (summary, chapters, tags, SEO, flashcards, quiz, timestamps, content DNA, viral score, revenue prediction).
- **video_reports** — User-submitted reports with category, description, and resolution status.

B.6 Premium and Payment Tables

- **premium_subscriptions** — Active subscription records with tier, gateway, transaction ID, start/end dates, and auto-renew flag.
- **payment_history** — Auditable log of all payment transactions across four gateways.

B.7 Scheduling and Feedback Tables

- **scheduled_videos** — Videos scheduled for future publication with target timestamp.
- **user_feedback** — Platform-wide feedback collection with category and priority fields.
- **video_feedback** — Per-video quality and content feedback from viewers.

Appendix C: AI Function Catalogue

Table 6.1 lists all 22 AI-powered functions implemented in the file `src/lib/ai.ts` (1,792 lines).

Table 6.1: *Complete AI Function Catalogue*

#	Function	Input	Output
1	<code>generateTitle</code>	Topic, keywords	SEO-optimized video title
2	<code>generateDescription</code>	Title, transcript	Structured description with timestamps
3	<code>generateTags</code>	Title, description	Ranked keyword tags
4	<code>generateSEO</code>	Video metadata	Meta description, Open Graph text
5	<code>generateSummary</code>	Transcript	Concise content summary
6	<code>generateChapters</code>	Transcript, duration	Timestamped chapter markers
7	<code>generateFlashcards</code>	Transcript	Q&A flashcard pairs
8	<code>generateQuiz</code>	Transcript	Multiple-choice quiz questions
9	<code>generateTimestamps</code>	Transcript	Key-moment timestamps
10	<code>generateContentDNA</code>	All metadata	12-dimension content fingerprint
11	<code>predictViralScore</code>	Metadata, trends	Virality probability (0–100)
12	<code>predictRevenue</code>	Views, engagement	Revenue estimate (NPR/USD)
13	<code>generateScriptCoach</code>	Draft script	Pacing and engagement feedback
14	<code>analyzeComment</code>	Comment text	Toxicity score, sentiment
15	<code>moderateContent</code>	Text content	Category, action recommendation
16	<code>detectSpam</code>	Comment text	Spam probability
17	<code>analyzeVideoContent</code>	Frame data, audio	NSFW score, violence score
18	<code>generateReply</code>	Comment, context	Contextual reply suggestion
19	<code>translateText</code>	Text, target lang	Translated text
20	<code>runViralSimulation</code>	Video metadata	Growth trajectory simulation
21	<code>transcribeAudio</code>	Audio URL	Text transcript (Whisper v3)
22	<code>generateSubtitles</code>	Transcript	WebVTT subtitle file

Appendix D: API Endpoint Reference

D.1 tRPC Sub-Routers (15 routers)

Table 6.2: *tRPC Router Summary*

Router	Procedures	Key Operations
videos	12	CRUD, search, trending, category filter, recommendation feed
comments	8	Create, list (threaded), react, pin, moderate, delete
users	7	Profile CRUD, role management, ban/unban, avatar update
subscriptions	5	Subscribe, unsubscribe, list subscribers, notification toggle
playlists	6	CRUD, add/remove videos, reorder
categories	3	List, create, delete
search	4	Global search, autocomplete, trending searches
notifications	4	List, mark read, mark all read, delete
watchHistory	3	Add entry, list, clear
community	8	Post CRUD, comment, like, poll vote
ai	10	All generation endpoints, moderation, transcription
premium	6	Tier info, subscribe, cancel, payment verify, history
admin	8	Dashboard stats, user management, content moderation queue
feedback	4	Submit, list, respond, analytics
reports	4	Submit, list, resolve, statistics

D.2 REST API Endpoints

- `POST /api/webhooks` — Clerk/Svix webhook handler for user synchronization

- `POST /api/uploadthing` — UploadThing file upload handler (video, thumbnail, and image)
- `GET /api/trpc/*` — tRPC HTTP handler (query operations)
- `POST /api/trpc/*` — tRPC HTTP handler (mutation operations)
- `GET /api/videos/[id]/stream` — Video streaming with range request support
- `POST /api/videos/[id]/transcode` — Video transcoding trigger
- `POST /api/payment/esewa/verify` — eSewa payment verification
- `POST /api/payment/khalti/verify` — Khalti payment verification
- `POST /api/payment/stripe/webhook` — Stripe webhook handler
- `POST /api/payment/paypal/webhook` — PayPal webhook handler

Appendix E: Technology Version Matrix

Table 6.3: *Complete Technology Stack with Versions*

Technology	Version	Purpose
Next.js	16.0.0	Full-stack React framework with App Router
React	19.0.0	UI component library
TypeScript	5.x	Static type checking
Tailwind CSS	3.4.x	Utility-first CSS framework
Drizzle ORM	0.39.x	Type-safe SQL query builder
PostgreSQL	16.x	Primary relational database
Neon	—	Serverless PostgreSQL hosting
Clerk	6.36.x	Authentication and user management
tRPC	11.0.x	End-to-end type-safe API layer
Bun	1.x	JavaScript/TypeScript runtime
UploadThing	7.7.x	File upload service
Radix UI	Various	Accessible UI primitives (48 components)
Lucide React	0.4.x	Icon library
Zustand	5.0.x	Lightweight state management
React Hook Form	7.x	Form state management
Zod	3.24.x	Schema validation
TanStack Query	5.x	Server state management (via tRPC)
Sonner	1.7.x	Toast notifications
date-fns	4.1.x	Date utility functions
Svix	1.x	Webhook verification
Pollinations AI	API	NLP text generation (18+ functions)
Replicate	API	Whisper v3 transcription, NSFW detection

Appendix F: Environment Configuration

The following environment variables must be configured for NepTube deployment:

```
# Database
DATABASE_URL=postgresql://user:pass@host/neptune

# Authentication (Clerk)
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=pk_...
CLERK_SECRET_KEY=sk_...
CLERK_WEBHOOK_SECRET=whsec_...

# File Upload (UploadThing)
UPLOADTHING_TOKEN=...

# AI Services
REPLICATE_API_TOKEN=...

# Payment Gateways
ESEWA_MERCHANT_ID=...
KHALTI_SECRET_KEY=...
STRIPE_SECRET_KEY=sk_...
STRIPE_WEBHOOK_SECRET=whsec_...
PAYPAL_CLIENT_ID=...
PAYPAL_CLIENT_SECRET=...

# Application
NEXT_PUBLIC_APP_URL=http://localhost:3000
VIDEO_SERVER_PORT=4000
```